

Subject: 23CSE301

Lab Session: 06

Notes & Coding Instructions:

1. Follow the instructions carefully given in this assignment.
2. You are encouraged to use AI tools for code generation and comparison.
3. The prompt log of your interaction should be generated from the tool and submitted.
4. While using GenAI tools, make sure you are doing it from your own login. We shall perform an audit of your interactions with the GenAI tools used.

Main Section (Mandatory):

A1. Please repeat the experiments conducted in Lab 05 (previous lab session) using AI tools. You may use GenAI tools such as Claude, ChatGPT, Co-Pilot. The function definitions and code modularization should be your own work and the AI tools should be used for code generation to meet your needs. Mention the GenAI tool used for each portion of code as comments in the code.

A2. Generate unit test cases using AI tools for testing your modular functions. The unit test cases should be used for both today's and last week's lab exercises.

A3. Conduct a performance comparison of the following 3 versions of k-NN algorithm.

- K-NN algorithm with Code written by you
- Using code for calling inbuilt functions of Scikit-Learn Package
- K-NN algorithm code developed using GenAI tool.

The performance comparison should be tabulated into your report with following evaluations: (i) accuracy metrics (Accuracy, Precision, Recall, F-score), and (ii) computational time calculated by averaging the time from 10 runs.

The test should be done on your project data.

Report Assignment:

1. Update your last week's report with the experiments and results obtained in this week's assignments.
2. The report should provide a list of unit and functional test cases executed and their outcomes.
3. Submit your GenAI tool Chat/prompt log as a pdf file (downloaded from the tool) to Teams assignment. The report should be submitted to TurnItIn.